Spectral Causal Theory (SCT): Research Roadmap and Task Prioritization

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1 What Has Been Proven

This section summarizes the results established in Part VII of the theory development. All results below are derived, not postulated.

1.1 Classical limit: Einstein–Hilbert action from spectral action

The spectral action principle

$$S_{\text{spec}} = \text{Tr}\left(f\left(\frac{D^2}{\Lambda^2}\right)\right) \tag{1}$$

where D is the Dirac operator of the spectral triple $(\mathcal{A}, \mathcal{H}, D)$ and f is a smooth cutoff function, reproduces in the IR limit the gravitational action:

$$S_{\text{spec}} \xrightarrow{\text{IR}} \frac{1}{16\pi G}\int d^4x\,\sqrt{g}\,(-R+2\Lambda_{\text{cc}}) + c_1\int d^4x\,\sqrt{g}\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + c_2\int d^4x\,\sqrt{g}\,R^2 + \cdots \tag{2}$$

with the cosmological constant Λ_{cc} and curvature-squared coefficients c_1, c_2 determined by moments of f.

Status: Proven. The coefficients are fixed by the spectral triple, not free parameters.

1.2 Quantum potential and the coefficient β

The coefficient in the one-loop effective potential:

$$\beta = \frac{41}{10\pi} \tag{3}$$

is reproduced automatically because the IR limit of SCT matches GR. This is a consistency check: SCT does not introduce new free parameters at one loop but reproduces the known result from the spectral structure.

Status: Proven as consequence of IR matching.

1.3 First unique prediction: the ratio c_1/c_2

The ratio of curvature-squared coefficients:

$$\frac{c_1}{c_2} = -\frac{1}{3} \tag{4}$$

is *fixed* by the spectral triple. In generic higher-derivative gravity, c_1 and c_2 are independent parameters. SCT predicts a definite relation, which is in principle testable via:

- gravitational wave dispersion in the high-frequency regime,
- modifications to black hole quasinormal mode spectra,
- corrections to Newtonian potential at short distances.

Status: Proven for the minimal spectral triple. **Caveat:** the sign correction $E = -R/4$ for Dirac fields (established in NT-1) changes $\hat{P} = -5R/12$ and gives $\beta_R^{\text{Dirac}} = 0$. The ratio c_1/c_2 for the full SM spectrum (with $N_s = 4$, $N_f = 45$, $N_v = 12$) must be re-derived in NT-1b, step (e). The value $-1/3$ may shift.

1.4 Nonlocal form factors: the master function

All five universal form factors arising in the one-loop gravitational effective action are generated by a single master function:

$$f(\xi) = \int_0^1 d\alpha \; e^{-\alpha(1-\alpha)\,\xi} \tag{5}$$

The five form factors $F_1, \ldots, F_5$ are expressible as combinations of $f(\xi)$ and its derivatives. The structure mirrors Barvinsky–Vilkovisky nonlocal heat kernel technology but here arises from spectral data rather than being postulated.

Status: Proven. Mathematical infrastructure established.

2 Near-Term Tasks

These tasks are the immediate priorities. Each builds directly on the proven results of Section 1.

2.1 NT-1: Explicit computation of $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2)$

Task: Compute the exact (non-expanded) form factors $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2)$ from the full spectral action (1), without truncating to the asymptotic heat kernel expansion.
Method: Two complementary approaches:

(a) **Zeta-function method.** Express the spectral action via the spectral zeta function $\zeta_D(s) = \mathrm{Tr}(|D|^{-2s})$. The form factors are obtained from the analytic continuation:

$$F_k(z) = \frac{1}{\Gamma(s_k)} \int_0^\infty dt \; t^{s_k - 1} \, \tilde{f}_k(t) \, e^{-t\,z}, \quad z = \Box/\Lambda^2 \tag{6}$$

where $\tilde{f}_k$ are determined by the cutoff function f and the Seeley–DeWitt coefficients of D^2.

(b) **Nonlocal heat kernel.** Use the Barvinsky–Vilkovisky technique extended to the full (resummed) heat kernel:

$$\mathrm{Tr}\left(f\left(\frac{D^2}{\Lambda^2} \right) \right) = \int_0^\infty ds \; \hat{f}(s) \; \mathrm{Tr}\left(e^{-sD^2/\Lambda^2} \right) \tag{7}$$

with $\hat{f}$ the inverse Laplace transform of f. Resum the curvature expansion to obtain closed-form F_1, F_2.

Expected outcome: Explicit analytic expressions for $F_1(z)$ and $F_2(z)$ valid for all $z \geq 0$, including:

- IR limit ($z \to 0$): must reproduce Eq. (2) with ratio (4).
- UV behavior ($z \to \infty$): determines whether the theory is asymptotically free, safe, or finite.
- Analyticity domain in the complex z-plane.

Dependencies: None (builds on proven results). This is the highest-priority task.

Status (2026-03-11): COMPLETE. Form factors derived for the massless Dirac operator. Seven rounds of independent audit (274+ quantitative checks, all PASS). Deep verification via seven-stage review pipeline with 95–98% confidence. See `theory/derivations/NT1_form_f`

2.2 NT-1b: Form factors for scalar and vector fields

Task: Extend the NT-1 computation to scalar (minimal and conformal coupling) and vector (gauge) fields, completing the full Standard Model spectrum.
Method: The CZ form factors [**?**] are universal for any Laplace-type operator $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$. Only the endomorphism E and connection curvature $\Omega_{\mu\nu}$ differ between field types:

Field	E	$\Omega_{\mu\nu}$	**tr(Id)**
Scalar ($\xi = 0$)	0	0	1
Scalar ($\xi = 1/6$)	$-R/6$	0	1
Dirac (massless)	$-R/4$	$R_{\mu\nu\rho\sigma}\gamma^\rho\gamma^\sigma/4$	4
Vector (gauge)	$R_{\mu\nu}$	$R_{\mu\nu\rho\sigma}$	d

(a) For each field type, compute the Weyl-basis form factors $h_C^{(s)}(x)$ and $h_R^{(s)}(x)$ using the CZ combination formulas with the appropriate traces.

(b) Verify local limits against known β-function coefficients:

$$\beta_W^{\text{scalar}} = \frac{1}{120}, \quad \beta_W^{\text{Dirac}} = \frac{1}{20}, \quad \beta_W^{\text{vector}} = \frac{1}{10} \tag{8}$$

(c) Construct the total form factors for the SM spectrum:

$$F_i^{\text{total}}(z) = N_s\, F_i^{\text{scalar}}(z) + N_f\, F_i^{\text{Dirac}}(z) + N_v\, F_i^{\text{vector}}(z) \tag{9}$$

with $N_s = 4$, $N_f = 45$, $N_v = 12$.

(d) Compute the total propagator and verify that the UV behavior of the combined form factors is consistent with the individual-field results.

(e) **Ratio c_1/c_2 re-derivation.** Compute the total local limits $c_i^{\text{total}} = N_s\, F_i^{\text{scalar}}(0) + N_f\, F_i^{\text{Dirac}}(0) + N_v\, F_i^{\text{vector}}(0)$ and determine the corrected ratio $c_1^{\text{total}}/c_2^{\text{total}}$. **Critical:** the sign correction $E = -R/4$ for Dirac (established in NT-1) gives $\hat{P} = -5R/12$ and $h_R(0) = 0$ ($\beta_R^{\text{Dirac}} = 0$). This changes the relative weight of Dirac vs. scalar/vector contributions and may shift the ratio from the original $-1/3$. If the corrected ratio differs from $-1/3$, update Section 1 and the prediction file `prediction_01_c1c2_ratio`.

Expected outcome:

- Explicit $h_C^{(s)}$, $h_R^{(s)}$ for scalar and vector fields.
- Total SM form factors $F_i^{\text{total}}(z)$ ready for use in MR-2 (unitarity), MR-3 (causality), and MR-7 (scattering amplitudes).
- Cross-check: local limits reproduce known one-loop β-functions for the full SM.
- **Updated c_1/c_2 ratio prediction** with corrected sign conventions, ready for experimental confrontation.

Dependencies: NT-1 (methodology and conventions established). Uses the same CZ form factor framework; only traces change.

Status (2026-03-11): Phases 1–2 COMPLETE.

- **Phase 1 (scalar):** $h_C^{(0)}$, $h_R^{(0)}(\xi)$ derived. $\beta_W^{(0)} = 1/120$, $\beta_R^{(0)} = \frac{1}{2}(\xi - 1/6)^2$. 142/142 quantitative checks PASS (4 errors caught by dual-review pipeline).
- **Phase 2 (vector):** $h_C^{(1)}$, $h_R^{(1)}$ derived. $\beta_W^{(1)} = 1/10$, $\beta_R^{(1)} = 0$. Ghost counting corrected: 2 Faddeev–Popov ghosts (not 1), shifting $\beta_W^{(1)}$ from $13/120 \to 1/10$. 1519 numerical checks ALL PASS (11 errors caught).
- **Phase 3 (combined SM):** Re-derivation of c_1/c_2 with $N_s = 4$, $N_f = 45$, $N_v = 12$ and corrected conventions—**PENDING**.

2.3 NT-2: Entire-function property of spectral form factors

Task: Determine whether the form factors $F_k(\Box/\Lambda^2)$ derived from the spectral action are entire functions of $\Box$, which would explain ghost-free behavior without additional postulates.

Method:

(a) From the explicit results of Task NT-1, analyze the singularity structure of $F_k(z)$ in the complex plane.

(b) Verify the Hadamard factorization: if F_k is entire of finite order ρ, then

$$F_k(z) = z^m \, e^{P(z)} \prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right) \tag{10}$$

with P a polynomial of degree $\leq \rho$ and E_p canonical factors. For ghost-freedom, there must be no real positive zeros (no poles in the propagator).

(c) Compare with postulated entire functions in ghost-free gravity (Tomboulis, Modesto, Biswas–Mazumdar–Siegel):

$$G(k^2) \sim \frac{e^{-\ell(k^2/\Lambda^2)}}{k^2 - m^2} \tag{11}$$

where ℓ is an entire function. Determine whether the spectral action *derives* this structure.

Expected outcome: Either:

(A) F_k are entire $\Rightarrow$ SCT is ghost-free *from first principles*, a major structural advantage over competing approaches.

(B) F_k have isolated poles $\Rightarrow$ identify their physical meaning (massive ghost modes, Lee–Wick partners, or artifacts of truncation).

Dependencies: Requires NT-1 (explicit form factors).

2.4 NT-3: Spectral dimension flow d_S(scale)

Task: Derive the running spectral dimension $d_S(\ell)$ as a function of the probing scale ℓ from the spectral properties of the Dirac operator D.
Method: The spectral dimension is defined via the return probability of a diffusion process:

$$d_S(\sigma) = -2\,\frac{d\ln P(\sigma)}{d\ln\sigma}, \quad P(\sigma) = \mathrm{Tr}\Big(e^{-\sigma D^2}\Big) \tag{12}$$

where σ is the diffusion time (inverse scale squared).

(a) Compute $P(\sigma)$ from the full spectrum of D^2 on the spectral triple.

(b) Expand for large σ (IR, should give $d_S \to 4$) and small σ (UV, test for dimensional reduction $d_S \to 2$).

(c) The $4 \to 2$ flow is a universal prediction of many quantum gravity approaches (CDT, Hořava–Lifshitz, asymptotic safety). Determine whether SCT reproduces this and with what specific flow profile.

Expected outcome: An explicit function $d_S(\sigma)$ interpolating between:

- $d_S \to 4$ at large scales (macroscopic spacetime),
- $d_S \to d_{\mathrm{UV}}$ at Planckian scales, with d_{UV} determined by the spectral triple.

If $d_{\mathrm{UV}} = 2$, this connects SCT to the dimensional reduction universality class. If $d_{\mathrm{UV}} \neq 2$, this is a distinguishing prediction.
Dependencies: Requires the spectral triple to be fully specified (Postulate 5 variants, see MT-3). A partial result is possible using the minimal spectral triple.

2.5 NT-4: Nonlocal gravitational field equations

Rationale for splitting. NT-4 is the single largest bottleneck in the roadmap: 17 downstream tasks depend on it directly or indirectly. The full nonlinear variational problem on arbitrary backgrounds is substantially harder than the linearized version on flat spacetime, yet many downstream tasks (MR-2, MR-3, MR-7, PPN-1) only require the linearized equations. Splitting NT-4 into three sub-tasks de-risks the critical path and enables earlier progress on physical predictions.

2.5.1 NT-4a: Linearized nonlocal equations on flat background

Task: Linearize the spectral action around flat spacetime $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and derive the linearized field equations in transverse-traceless gauge.
Method:

(a) Expand the spectral action (14) to second order in $h_{\mu\nu}$. On flat background, $\Box \to \partial^2$ and the form factors become ordinary functions of the d'Alembertian: $F_i(\partial^2/\Lambda^2)$.

(b) Extract the linearized equations in transverse-traceless gauge. The result is a modified wave equation:

$$\mathcal{G}^{(1)}_{\mu\nu}[h] + \big[\alpha_C\, F_1(\partial^2/\Lambda^2) + \alpha_R\, F_2(\partial^2/\Lambda^2)\big]\, \mathcal{G}^{(1)}_{\mu\nu}[h] = 8\pi G\, T^{(1)}_{\mu\nu} \tag{13}$$

where the coefficients $\alpha_{C,R}$ arise from the linearized Weyl and Ricci contributions.

(c) Verify: (i) the equations reduce to linearized Einstein equations when $F_1, F_2 \to c_1, c_2$; (ii) gauge invariance under $h_{\mu\nu} \to h_{\mu\nu} + \nabla_{(\mu}\xi_{\nu)}$; (iii) the propagator derived from these equations matches MR-2.

(d) Read off the modified Newtonian potential for PPN-1 analysis.

Expected outcome:

- Linearized nonlocal field equations ready for propagator analysis (MR-2), causality analysis (MR-3), and PPN computation (PPN-1).
- Modified Newtonian potential as a function of Λ and F_i.
- Scattering amplitude input for MR-7.

Dependencies: NT-1 (Dirac form factors), NT-1b (total form factors for the physical field equations with correct SM spectrum).

2.5.2 NT-4b: Full nonlinear variational equations

Task: Derive the full nonlinear nonlocal gravitational field equations from variation of the spectral action on arbitrary backgrounds:

$$G_{\mu\nu} + 2\,\frac{\delta}{\delta g^{\mu\nu}} \int d^4x\,\sqrt{g}\,\left[F_1(\Box/\Lambda^2)\,C_{\alpha\beta\gamma\delta}C^{\alpha\beta\gamma\delta} + F_2(\Box/\Lambda^2)\,R^2\right] = 8\pi G\,T_{\mu\nu} \tag{14}$$

The nontrivial part is the variational calculus of the nonlocal operators $F_i(\Box)\,\mathcal{R}^2$ with respect to the metric on curved backgrounds.

Method:

(a) Use the Barvinsky–Vilkovisky technique for variation of nonlocal functionals. The key identity is:

$$\frac{\delta}{\delta g^{\mu\nu}}\left[A\,F(\Box)\,B\right] = \int_0^1 d\alpha\,\left[F'(\Box_\alpha)\,(\delta\Box)\right]_{\alpha\text{-insertion}} + \text{boundary terms} \tag{15}$$

where the α-insertion accounts for the non-commutativity of $\delta\Box$ with functions of $\Box$.

(b) Specialize to the Weyl tensor and Ricci scalar terms separately. For the R^2 sector, the variation gives $\nabla_\mu\nabla_\nu[F_2(\Box)R]$ terms; for C^2, the variation involves the Bach tensor and its nonlocal generalization.

(c) Verify: (i) the equations reduce to Einstein's equations when $F_1, F_2 \to c_1, c_2$ (local limit); (ii) the trace equation is consistent; (iii) the Bianchi identity $\nabla^\mu(\text{l.h.s.}) = 0$ is satisfied; (iv) the linearized limit on flat background reproduces NT-4a.

(d) Derive the explicit Wald entropy variation $\delta S_{\text{spec}}/\delta R_{\mu\nu\rho\sigma}$ for MT-1.

Expected outcome:

- Closed-form nonlinear nonlocal field equations on arbitrary backgrounds.
- Wald entropy variation for MT-1.
- Foundation for MR-9 (black hole singularity resolution on Schwarzschild background).

Dependencies: NT-4a (linearized limit as cross-check), NT-1, NT-1b.

2.5.3 NT-4c: FLRW reduction of nonlocal equations

Task: Reduce the full nonlocal field equations (NT-4b) to the FLRW metric $ds^2 = -dt^2 + a(t)^2\, d\vec{x}^2$ and derive the modified Friedmann equations including nonlocal form factor contributions.
Method:

(a) Substitute the FLRW metric into the full nonlocal equations of NT-4b. On FLRW backgrounds, $\Box = -\partial_t^2 - 3H\partial_t$ (for scalar quantities), which simplifies the form factors to functions of the conformal time derivative.

(b) Extract the modified Friedmann equations: $H^2 + \delta H^2_{\rm NL}(F_1, F_2) = \frac{8\pi G}{3}\rho$, where $\delta H^2_{\rm NL}$ encodes the nonlocal corrections.

(c) Compute the inflationary perturbation equations to first order in $\delta g_{\mu\nu}$ on FLRW background (needed for n_s, r in INF-1).

(d) Verify: (i) GR limit ($F_i \to c_i$) gives standard Friedmann; (ii) de Sitter solution exists; (iii) consistency with energy conservation $\dot{\rho} + 3H(\rho + p) = 0$ (possibly modified).

Expected outcome:

- Modified Friedmann equations for cosmological applications (MT-2, INF-1, LT-3c).
- Perturbation equations on FLRW for inflationary observables.
- De Sitter stability analysis.

Dependencies: NT-4b (full nonlinear equations), NT-1b (total form factors).

2.6 META-1: Parameter counting and falsifiability criteria

Task: Perform a systematic parameter count of SCT and establish explicit falsifiability criteria for each prediction channel.
Method:

(a) **Free parameter census.** Enumerate all free parameters of SCT: the spectral function f (or equivalently its moments f_0, f_2, f_4), the finite Dirac operator D_F (13 Yukawa parameters + Majorana masses), and the noncommutativity scale Λ. Determine which are independently measurable and which are fixed by internal relations (e.g., sum rules).

(b) **Comparison with competitors.** Tabulate the parameter count against: ΛCDM (6), SM (19), string theory landscape ($\sim 10^{500}$ vacua), LQG (Immirzi γ), asymptotic safety (relevant couplings at fixed point).

(c) **Falsifiability matrix.** For each prediction (QNM shifts, $w(z)$, $\Delta F/F$, PPN parameters, c_1/c_2, inflationary observables, spectral dimension), specify: (i) the predicted value or range as a function of Λ, (ii) the required experimental precision, (iii) the dataset that tests it, (iv) what outcome would *falsify* SCT (not merely constrain Λ).

(d) **Theory review checklist.** Address all 10 mandatory checks from the physics methodology: degrees of freedom, action, symmetries, observables, domain of validity, limiting cases, predictions, falsification, novelty, rigor vs. heuristics.

(e) **Cutoff function f analysis (structural gap).** The spectral action $\mathrm{Tr}(f(D^2/\Lambda^2))$ depends on the cutoff function f, which is an *infinite-dimensional* free input not fixed by any postulate. This is the most serious structural ambiguity in the spectral action program. Systematically analyze:

(i) **Sensitivity audit:** Which predictions depend only on the moments f_0, f_2, f_4 (i.e., are universal), and which depend on the full shape of f? The heat-kernel expansion to a_4 (Einstein–Hilbert + SM Lagrangian) depends only on moments; the nonlocal form factors F_1, F_2 depend on the *entire* function f via the Laplace transform.

(ii) **Classification:** Partition all roadmap predictions into "moment-dependent" (robust) vs. "f-shape-dependent" (fragile). Specifically: $c_1/c_2 = -1/3$ is moment-independent (universal ratio from heat kernel); the form factors $F_i(z)$ and their UV behavior *are* f-dependent.

(iii) **Constraints on f:** Determine whether physical requirements (unitarity via MR-2, causality via MR-3, finiteness via MR-4/MR-5, entire-function property via NT-2) constrain f sufficiently to reduce the ambiguity from infinite-dimensional to finite-dimensional.

(iv) **Universality test:** For predictions that do depend on f, determine whether the *qualitative* predictions (sign, order of magnitude, asymptotic behavior) are stable under variation of f within the allowed class, or whether different choices of f can produce qualitatively different physics.

This analysis is essential for honestly assessing which SCT predictions are genuine and which are artifacts of a particular f-choice.

Expected outcome:

- Complete parameter table with comparison to competing approaches.
- Falsifiability matrix ready for publication.
- Self-assessment document identifying what is rigorous, what is heuristic, and what remains open in SCT.
- Cutoff function sensitivity report classifying all predictions as universal (moment-only) vs. f-dependent, with explicit error bars from f-variation where applicable.

Dependencies: None (meta-task; can proceed immediately). Should be updated as each prediction task completes.

3 Mid-Term Tasks

These tasks extend SCT to key physical domains. Each requires results from at least one near-term task.

3.1 MT-1: Black hole entropy from spectral triple

Task: Derive the Bekenstein–Hawking entropy $S_{\mathrm{BH}} = A/(4G)$ from the spectral triple, and compute leading quantum corrections.
Method:

(a) **Entanglement entropy approach.** For a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ restricted to a causal diamond bounded by a horizon, compute the entanglement entropy of the Dirac vacuum:

$$S_{\mathrm{ent}} = -\operatorname{Tr}(\rho_{\mathrm{in}} \ln \rho_{\mathrm{in}}), \quad \rho_{\mathrm{in}} = \operatorname{Tr}_{\mathrm{out}} |\Omega\rangle\langle\Omega| \tag{16}$$

(b) **Spectral action on horizon.** Evaluate the spectral action on the near-horizon geometry (Rindler approximation) and identify the entropy functional:

$$S = -2\pi \int_{\mathrm{horizon}} \frac{\delta S_{\mathrm{spec}}}{\delta R_{\mu\nu\rho\sigma}} \epsilon_{\mu\nu}\epsilon_{\rho\sigma}\, d\Sigma \tag{17}$$

(Wald entropy formula applied to the spectral action).

(c) **Logarithmic corrections.** The spectral action naturally includes higher-curvature terms. Compute the correction:

$$S_{\mathrm{BH}} = \frac{A}{4G} + c_{\log} \ln\left(\frac{A}{\ell_P^2}\right) + \cdots \tag{18}$$

and determine $c_{\log}$ from the spectral data. Compare with loop quantum gravity ($c_{\log} = -3/2$ for BTZ) and string theory results.

Expected outcome:

- Derivation of area law from spectral principles.
- Predicted value of $c_{\log}$ as a testable output.
- Understanding of how the spectral triple encodes horizon degrees of freedom.
- **All four laws of BH thermodynamics:** zeroth law (constancy of surface gravity on horizon), first law (energy balance $dM = (\kappa/8\pi G)\, dA + \ldots$ with spectral corrections), second law (area non-decrease or generalized second law with quantum corrections), third law (extremal limit). Note: a 2025 result (arXiv:2509.23423) shows the conventional first law is violated in NCG because entropy deviates from Bekenstein–Hawking.
- **Hawking radiation spectrum.** Derive the grey-body factors from the nonlocal effective potential around the black hole. The nonlocal form factors modify the barrier in the Regge–Wheeler/Zerilli equations, potentially altering the emission spectrum at frequencies $\omega \sim \Lambda$.

Dependencies: NT-1 (explicit form factors for Wald entropy computation), NT-1b (total SM form factors for Wald entropy with correct field content), NT-4b (full nonlinear field equations—variational Wald entropy derivation requires the complete nonlinear equations, not just the linearized sector).

Note: NT-3 (spectral dimension) was removed as a dependency. The near-horizon spectral dimension is an *output* of MT-1 (computed from the Wald entropy result), not an input. The spectral dimension enters the *interpretation* of the entropy result, not its derivation.

3.2 MT-2: Modified Friedmann equations from SCT

Task: Derive the cosmological evolution equations from the spectral action applied to the Friedmann–Lemaître–Robertson–Walker (FLRW) metric.
Method:

(a) Evaluate the spectral action (1) on the FLRW metric $ds^2 = -dt^2 + a(t)^2\, d\vec{x}^2$ with the standard Dirac operator.

(b) Variation with respect to $a(t)$ yields modified Friedmann equations:

$$H^2 = \frac{8\pi G}{3}\,\rho + \delta_1(H/\Lambda) + \delta_2(H/\Lambda) \tag{19}$$

where δ_1, δ_2 are corrections from the nonlocal form factors.

(c) Compute the effective equation of state $w_{\text{eff}}(z)$ and compare with ΛCDM and observational data (Planck 2018, DESI DR1/DR2 BAO, Pantheon+ SNe, ACT DR6).

(d) Analyze the early-universe regime $H \sim \Lambda$: determine whether SCT predicts a bounce, inflationary phase, or other modification.

Expected outcome:

- Explicit modified Friedmann equations with SCT corrections.
- Predicted deviation $w(z) - w_{\Lambda\text{CDM}}(z)$ as function of redshift.
- Constraints on Λ from CMB and BAO data.
- Determination of early-universe behavior (bounce vs. inflation vs. other).
- **Hubble tension benchmark:** Compute H_0 from SCT-modified expansion history and compare with Planck (indirect, 67.4 ± 0.5) and SH0ES (direct, 73.0 ± 1.0). Maggiore's nonlocal RR model reduces H_0 tension from $> 5\sigma$ to $\sim 2.0\sigma$ by predicting phantom-like $w_{\text{DE}} \approx -1.14$; SCT should benchmark against this.
- S_8 **tension benchmark:** Compute σ_8 from SCT-modified growth function $f\sigma_8(z)$ via the modified Poisson equation. Nonlocal gravity generically lowers the growth rate, helping alleviate the S_8 tension (weak lensing vs. CMB: KiDS, DES, HSC constraints).
- **Cosmological constant consistency check:** Determine how nonlocal form factors modify the effective cosmological constant $\Lambda_{\text{CC}}^{\text{eff}}$ relative to the bare spectral action value $f_0\Lambda^4/(16\pi^2)$. Specifically, check whether the nonlocal dressing suppresses the bare Λ^4 contribution (as suggested by the spectral action's insensitivity to additive vacuum energy shifts). Compare the resulting $\Lambda_{\text{CC}}^{\text{eff}}$ with the causal set prediction $\Lambda \sim 1/\sqrt{N}$ (Sorkin's "everpresent Λ"). This is an internal consistency test: two components of SCT (spectral action and causal set sector) predict the same observable via different mechanisms, and they must agree or one must dominate.

Dependencies: NT-1 (Dirac form factors), NT-1b (total SM form factors—the FLRW corrections δ_1, δ_2 in Eq. (19) depend on F_i^{total}, not Dirac alone), NT-2 (entire-function property determines whether cosmological perturbation theory is well-defined), NT-4c (FLRW reduction of the nonlocal field equations—the modified Friedmann equations are obtained by specializing to FRW symmetry).

3.3 MT-3: Three variants of Postulate 5

Task: Develop, compare, and select among three candidate formulations of Postulate 5 (the dynamical principle connecting spectral data to spacetime evolution).
Method: The three variants under consideration:

(V1) **Spectral action extremization.** Spacetime geometry is determined by extremizing the spectral action:

$$\delta \operatorname{Tr}\left(f\left(\frac{D^2}{\Lambda^2}\right)\right) = 0 \tag{20}$$

with D depending on the metric (and possibly matter fields) through the Dirac operator construction.

(V2) **Spectral flow dynamics.** The Dirac operator evolves via a spectral flow equation:

$$\frac{\partial D}{\partial t} = \mathcal{F}[D, \operatorname{Spec}(D)] \tag{21}$$

where $\mathcal{F}$ is a functional determined by the spectral data. This is analogous to Ricci flow but for the full spectral triple.

(V3) **Path integral over spectral triples.** The quantum theory is defined via a sum over spectral triples:

$$Z = \int \mathcal{D}[D]\, e^{-S_{\text{spec}}[D]} \tag{22}$$

with a suitable measure $\mathcal{D}[D]$ on the space of Dirac operators.

For each variant:

- Check mathematical well-posedness (existence, uniqueness, regularity).
- Verify that the classical limit gives GR.
- Determine what new predictions arise.
- Assess compatibility with unitarity and causality.

Expected outcome:

- Detailed comparison table of three variants.
- Identification of the most promising variant (or demonstration that they are equivalent in some regime).
- Mathematical framework for the chosen variant.

Dependencies: NT-1, NT-1b (total SM form factors—assessing well-posedness requires the physical form factors F_i^{total}, not just the Dirac sector), NT-2 (needed to assess whether the action principle is well-posed in each variant).

3.4 INF-1: Inflationary predictions from the spectral action

Task: Derive the inflationary observables (n_s, r, $f_{\rm NL}$, reheating temperature $T_{\rm reh}$) from the spectral action's Starobinsky-like R^2 sector, including the effects of the nonlocal form factors.

Method:

(a) **Inflationary potential.** The spectral action's R^2 term drives Starobinsky inflation. In the Einstein frame, the scalaron φ has the potential

$$V(\varphi) = \frac{3M^2}{4}\left(1 - e^{-\sqrt{2/3}\,\varphi/M_P}\right)^2 \tag{23}$$

with the scalaron mass $M^2 = \Lambda^2/(6c_2)$ determined by the spectral data. Compute M from the total $c_2^{\rm total}$ (NT-1b).

(b) **Nonlocal corrections.** The full spectral action replaces $c_2 R^2$ with $R\,F_2(\Box/\Lambda^2)\,R$. Following Koshelev–Kumar–Starobinsky (JHEP 2020), compute the modified slow-roll parameters and their effect on n_s, r. Standard Starobinsky: $n_s \approx 1 - 2/N \approx 0.965$, $r \approx 12/N^2 \approx 3.3\times10^{-3}$ for $N = 55$ e-folds. The nonlocal form factors modify r at the few-percent level.

(c) **Non-Gaussianity.** Compute $f_{\rm NL}$ from the three-point function of curvature perturbations. In single-field slow-roll, $f_{\rm NL} \sim \mathcal{O}(10^{-2})$; the additional NCG scalars (σ-field from seesaw sector, dilaton) could enhance it.

(d) **Reheating.** In the nonlocal theory, the scalaron's coupling to matter fields is modified by the form factors. Compute the decay rate Γ_φ and the resulting reheating temperature $T_{\rm reh}$. Verify consistency with BBN constraints ($T_{\rm reh} > 5$ MeV).

(e) **CMB confrontation.** Compare predictions with Planck 2018 ($n_s = 0.9649 \pm 0.0042$, $r < 0.036$), BICEP/Keck ($r < 0.036$), the upcoming Simons Observatory ($\sigma_r \approx 0.003$), and LiteBIRD sensitivity ($\sigma_r \approx 10^{-3}$).

(f) **De Sitter conjecture check.** The refined Swampland de Sitter conjecture (Ooguri–Palti–Shiu–Vafa 2018, arXiv:1810.05506) states $|\nabla V|/V \geq c \sim \mathcal{O}(1)$ or $\min(\nabla_i\nabla_j V) \leq -c'\,V$ in any consistent quantum gravity theory. If the spectral action's scalar potential (Einstein-frame scalaron + NCG Higgs sector) has a stable de Sitter vacuum, determine whether this violates the conjecture and what this implies for SCT's relationship to the Swampland program. If the potential *does* satisfy the conjecture (quintessence-like behavior), determine the constraints on the cutoff function f and Λ. This is a self-consistency check, not a Swampland endorsement—the point is to know where SCT stands on a concrete question that competing programs (string theory, AS) must also address.

Expected outcome:

- Predicted values of n_s, r, $f_{\rm NL}$, $T_{\rm reh}$ as functions of Λ and the spectral data.
- Assessment of whether SCT-Starobinsky inflation is within LiteBIRD detection range ($r \approx 4\times10^{-3}$ at $\sim 4\sigma$).
- Comparison with standard Starobinsky, α-attractor models, and nonlocal Starobinsky (Koshelev et al.).

Dependencies: NT-1b (total c_2^{total} for scalaron mass), NT-2 (analyticity of form factors for nonlocal slow-roll analysis), NT-4c (FRW reduction of nonlocal field equations for perturbation theory).

Note: MT-2 is *not* a prerequisite: INF-1 derives the inflationary dynamics *from* the spectral action directly (Einstein-frame scalaron potential), not from the modified Friedmann equations. MT-2 and INF-1 are parallel tracks that independently reduce the spectral action to cosmological settings.

4 Theory Consistency Tasks (MR-1 through MR-9)

These tasks address the fundamental consistency requirements for a UV-complete quantum gravity theory. A theory that fails any of the CRITICAL tasks below cannot claim UV-completeness regardless of other results.

4.1 MR-1: Lorentzian spectral action

Task: Define the spectral action in Lorentzian signature and resolve the tension between the Lorentzian causal structure (Postulate 2) and the Euclidean spectral action (Postulate 3).
Method:

(a) Investigate the **Krein space formulation** of spectral triples (Strohmaier, Paschke–Sitarz, van den Dungen–Rennie): replace the Hilbert space $\mathcal{H}$ with a Krein space $(\mathcal{K}, J)$ where J is a fundamental symmetry encoding the indefinite metric of Lorentzian signature.

(b) Define D^2 with Lorentzian signature and determine whether $\mathrm{Tr}(f(D^2/\Lambda^2))$ is well-defined (the operator D^2 is no longer positive-definite).

(c) Prove or disprove that the Wick rotation $z_E \to z_L = -z_E$ applied to the Euclidean form factors (NT-1) yields a valid Lorentzian action at *all* curvature scales, not just in the weak-field limit.

(d) Determine whether the causal set structure (Postulate 2) constrains the allowed Lorentzian spectral triples, providing a selection principle.

Expected outcome: Either a well-defined Lorentzian spectral action, or a precise characterization of the domain of validity of the Euclidean formulation.
Dependencies: None (foundational). Can proceed in parallel with all other tasks.
Severity: CRITICAL.

4.2 MR-2: Unitarity and stability

Task: Prove that the full graviton propagator in the nonlocal SCT action has no ghost poles (no negative-residue poles on the physical sheet), verify the optical theorem, and establish Hamiltonian stability.
Method:

(a) From the total form factors (NT-1b, Eq. (9)), construct the full momentum-space graviton propagator in the transverse-traceless decomposition:

$$G^{-1}_{\mu\nu\rho\sigma}(k^2) = G^{-1}_{\mathrm{EH}} + k^4\big[F_1^{\mathrm{tot}}(k^2/\Lambda^2)\,\mathcal{P}^{(2)} + F_2^{\mathrm{tot}}(k^2/\Lambda^2)\,\mathcal{P}^{(0)}\big] \tag{24}$$

(b) Analyze all poles: positions $k^2 = m_n^2$, residues Z_n, and their signs. Ghost-freedom requires $Z_n > 0$ for all physical poles.

(c) **Massive spin-2 mode (Stelle ghost).** In the local truncation ($F_1 \to c_1$), the Weyl-squared term produces a massive spin-2 ghost with $m^2 = 1/(2|c_1|)$. Determine whether the nonlocal form factors $F_1(z)$ cure this ghost (cf. Tomboulis, Modesto, Biswas–Mazumdar–Siegel).

(d) **Scalar mode.** The SCT prediction $3c_1 + c_2 = 0$ implies $m_0^2 \to \infty$ (scalar mode decouples). Verify this persists in the nonlocal theory: $3F_1(z)+F_2(z) \neq 0$ for $z > 0$?

(e) **Optical theorem.** Compute the imaginary part of the one-loop graviton self-energy and verify $\operatorname{Im}\Pi(s) \geq 0$ for $s > 0$.

(f) **Ostrogradski stability.** Analyze the Hamiltonian structure of the nonlocal equations of motion. For infinite-derivative theories, the Ostrogradski theorem does not apply directly, but stability of the initial value problem must be established.

(g) **Conformal factor problem.** Verify that the spectral action with SCT-fixed coefficients (c_1, c_2) is bounded below in the conformal sector, resolving the conformal factor instability of Euclidean quantum gravity.

(h) **Positive energy.** Re-examine the Schoen–Yau / Witten positive energy theorem for the nonlocal gravitational action.

Expected outcome: A definitive answer on ghost-freedom, unitarity, and stability of the SCT gravitational sector.
Dependencies: NT-1 (Dirac form factors), NT-1b (total form factors), NT-2 (entire-function property), NT-4a (linearized nonlocal field equations—ghost analysis requires the explicit linearized equations of motion).
Severity: CRITICAL.

4.3 MR-3: Causality in the nonlocal theory

Task: Prove that the nonlocal equations of motion from the spectral action have a well-posed initial value problem and do not produce acausal propagation.
Method:

(a) Analyze the **retarded Green's function** of the linearized nonlocal equations of motion. Verify that its support is contained in the forward light cone.

(b) Check **Kramers–Kronig relations** for the graviton self-energy:

$$\operatorname{Re}\Pi(\omega) = \frac{1}{\pi}\,\mathcal{P}\int_{-\infty}^{\infty} \frac{\operatorname{Im}\Pi(\omega')}{\omega' - \omega}\,d\omega' \tag{25}$$

(c) Compare with known results on causality in infinite-derivative gravity (Buoninfante, Modesto, Rachwal): classify the SCT form factors according to the causal/acausal taxonomy.

(d) Determine whether the **initial value problem** for the nonlocal field equations is well-posed in the sense of Hadamard (existence, uniqueness, continuous dependence on initial data).

(e) **Macrocausality (cluster decomposition).** Nonlocal theories with entire-function form factors violate strict cluster decomposition at distances $\sim 1/M$. Prove approximate factorization (macrocausality) at distances $\gg 1/M$ for the SCT action. Krasnikov (2024) established macrocausality for certain nonlocal ϕ^4 models; extend to the gravitational case.

Expected outcome: Proof of causality preservation or identification of acausal regimes and their physical interpretation. Proof of macrocausality at distances $\gg 1/\Lambda$.
Dependencies: NT-1 (form factors), NT-1b (total SM form factors—causality of the physical retarded propagator requires F_i^{total}), NT-2 (analyticity structure), NT-4a (linearized nonlocal field equations).
Severity: CRITICAL.

4.4 MR-6: Convergence of the curvature expansion

Task: Determine whether the curvature expansion of the spectral action converges, and if so, with what radius of convergence.
Method:

(a) Estimate the growth rate of Seeley–DeWitt coefficients a_{2k} using known results and combinatorial bounds.

(b) Analyze the Laplace transform representation $\mathrm{Tr}(f(D^2/\Lambda^2)) = \int_0^\infty \hat{f}(t)\ \mathrm{Tr}(e^{-tD^2/\Lambda^2})\, dt$ and determine whether it provides a rigorous non-perturbative definition.

(c) For specific spectral triples (e.g., round S^4 with radius a), compute the spectral action *exactly* from the known spectrum of D^2 and compare with the asymptotic expansion truncated at various orders.

(d) Determine the domain $\|R\|/\Lambda^2 < r_{\text{conv}}$ where the $\mathcal{O}(\mathcal{R}^2)$ truncation is reliable.

Expected outcome: Rigorous statement about the relationship between the perturbative (heat kernel) expansion and the exact spectral action. Radius of convergence or proof of asymptotic (non-convergent) nature.
Dependencies: None (can proceed in parallel with all other tasks).
Severity: IMPORTANT.

4.5 MR-4: Two-loop effective action

Task: Compute the two-loop contribution to the gravitational effective action from the spectral action and verify that no new divergences arise that are not absorbed by the spectral function ψ.
Method:

(a) Use the background field method with the nonlocal one-loop action (NT-1, NT-1b) as the starting point. Expand the metric as $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$ and compute the two-loop diagrams with nonlocal vertices.

(b) Identify the structure of two-loop divergences: are they of the form R^3, $\nabla^2 R^2$, or new nonlocal structures?

(c) Determine whether the spectral function $\psi(u)$ (entering via $f(D^2/\Lambda^2) = \int_0^\infty \psi(u)\, e^{-uD^2/\Lambda^2}\, du$) absorbs all two-loop divergences, or whether new counterterms are needed.

(d) Compare with known two-loop results in higher-derivative gravity (Goroff–Sagnotti for pure gravity, van de Ven for gravity + matter).

Expected outcome: Either: (A) all two-loop divergences are absorbed by $\psi \Rightarrow$ evidence for all-orders finiteness, or (B) new counterterms required $\Rightarrow$ identifies what additional structure SCT needs.
Dependencies: NT-1 (one-loop form factors), NT-1b (full SM spectrum), MR-2 (ghost-freedom needed to define loop expansion consistently), MR-7 (one-loop scattering amplitudes provide the integrand structure that the two-loop calculation must iterate).
Severity: CRITICAL.

4.6 MR-5: Power-counting and all-orders finiteness argument

Task: Establish a power-counting argument (or prove its impossibility) for the finiteness of the spectral action at all loop orders.
Method:

(a) Analyze the superficial degree of divergence D_γ for L-loop diagrams with nonlocal vertices from the spectral action. The nonlocal form factors modify the standard power-counting:

$$D_\gamma = 4 - E_{\text{ext}} + 2L - \sum_v \delta_v(k^2/\Lambda^2) \tag{26}$$

where δ_v encodes the UV suppression from form factors at vertex v.

(b) For Schwartz-class spectral functions ($\psi \in \mathcal{S}(\mathbb{R}_+)$), determine whether the exponential UV suppression renders all diagrams finite.

(c) For merely exponential spectral functions ($\psi = e^{-u}$), determine whether the $1/z$ UV decay of form factors is sufficient for finiteness or whether it fails at sufficiently high loop order.

(d) If an all-orders argument exists, formalize it as a theorem with precise conditions on ψ.

(e) Compare with known results: Tomboulis's argument for entire-function gravity, Krasnikov's super-renormalizability claims, and Modesto's power-counting for nonlocal gravity.

(f) **Naturalness and hierarchy problem.** If SCT is UV-finite, analyze whether the Higgs mass is protected from quadratic divergences (the hierarchy problem). The spectral action's sum rule $m_H^2 \propto \mathrm{Tr}(D_F^4) - \mathrm{Tr}(D_F^2)^2/\,\mathrm{Tr}(\mathrm{Id})$ fixes m_H at the unification scale; determine whether the nonlocal form factors stabilize this prediction under radiative corrections without fine-tuning.

Expected outcome: Either: (A) a theorem stating conditions on ψ for all-orders finiteness (the strongest UV-completeness result), or (B) identification of the loop order at which finiteness fails and what is needed to fix it. Additionally: assessment of naturalness—whether SCT radiative corrections preserve the Higgs mass prediction without fine-tuning.
Dependencies: NT-1 (form factor structure), NT-1b (total SM form factors—power-counting must account for full matter content and total UV behavior of vertices), NT-2 (entire-function property), MR-4 (two-loop as explicit check of any general argument).

Severity: CRITICAL.

4.7 MR-7: Scattering amplitudes and gauge-fixing

Task: Compute graviton–graviton scattering amplitudes from the nonlocal spectral action and establish a consistent gauge-fixing procedure for the nonlocal theory.
Method:

(a) **Gauge-fixing.** The nonlocal action requires a modified gauge-fixing procedure. The standard de Donder gauge $\mathcal{F}_\mu = \nabla^\nu h_{\mu\nu} - \frac{1}{2}\nabla_\mu h$ must be supplemented with nonlocal gauge-fixing terms:

$$S_{\text{gf}} = \frac{1}{2\alpha} \int d^4x \sqrt{g}\, \mathcal{F}_\mu\, \mathcal{G}^{\mu\nu}(\Box/\Lambda^2)\, \mathcal{F}_\nu \tag{27}$$

where $\mathcal{G}^{\mu\nu}$ is chosen to simplify the propagator. Determine the optimal choice and verify BRST invariance.

(b) **Ghost action.** Derive the Faddeev–Popov ghost action for the nonlocal gauge-fixing. Verify that ghosts do not introduce new physical poles.

(c) **Tree-level amplitudes.** Compute the $2 \to 2$ graviton scattering amplitude at tree level with the full nonlocal propagator. Verify that: (i) the amplitude reduces to GR in the IR, (ii) the UV behavior is improved, (iii) no acausal features appear.

(d) **One-loop amplitudes.** Compute the one-loop $2 \to 2$ amplitude and verify: (i) gauge invariance (Ward identities), (ii) unitarity (optical theorem), (iii) UV finiteness.

(e) **Matter sector unitarity.** Formulate and prove a theorem that the spectral invariance of $\mathrm{Tr}(f(D^2/\Lambda^2))$ automatically guarantees the Ward–Takahashi and Slavnov–Taylor identities for graviton–fermion and graviton–gauge boson scattering, ensuring that the nonlocal form factors do not spoil matter sector unitarity.

Expected outcome:

- Consistent gauge-fixed nonlocal action with BRST symmetry.
- Explicit graviton scattering amplitudes at tree and one-loop level.
- Verification of unitarity, gauge invariance, and improved UV behavior.

Dependencies: NT-1b (total form factors), NT-4a (linearized nonlocal field equations—needed for the external states in scattering amplitudes), MR-2 (ghost-freedom of propagator), MR-3 (causality of Green's functions).

Severity: IMPORTANT.

4.8 MR-8: Non-perturbative definition

Task: Define the quantum theory non-perturbatively via a path integral (or equivalent construction) over Dirac operators and establish the existence and properties of the measure $\mathcal{D}[D]$.
Method:

(a) Formalize the space $\mathcal{M}_D$ of Dirac operators compatible with the spectral triple axioms. Determine its topology and geometry.

(b) Define a measure $\mathcal{D}[D]$ on $\mathcal{M}_D$ that is: (i) diffeomorphism-invariant, (ii) positive-definite (Euclidean) or appropriately oscillatory (Lorentzian), (iii) compatible with the spectral action as the weight function.

(c) Investigate discretized approximations: the spectral action on finite-dimensional spectral triples (matrix geometries, fuzzy spaces) as regularized versions of the full path integral.

(d) Compare with: Causal Dynamical Triangulations (CDT), Euclidean Dynamical Triangulations (EDT), and random matrix models for noncommutative geometries (Barrett–Glaser).

(e) **Causal set convergence (background independence check).** For causal sets obtained by Poisson sprinkling on known manifolds (S^4, T^4, Schwarzschild), construct the truncated spectral triple (finite matrix D_N) following the Glaser–Stern method (2020–2021) and verify convergence of the spectral action $\mathrm{Tr}(f(D_N^2/\Lambda^2)) \to \mathrm{Tr}(f(D^2/\Lambda^2))$ as $N \to \infty$. This validates the assumption that SCT's continuum calculations (NT-1, NT-1b, NT-4a/b) are the correct large-N limit of the fundamental discrete structure postulated in Postulates 1–2.

(f) **Benincasa–Dowker action limit.** The Benincasa–Dowker (BD) action is the *only* known discrete action on causal sets that reproduces $\int R\sqrt{g}\, d^4x$ in the continuum limit (Benincasa–Dowker 2010, arXiv:1001.2725). Verify that the spectral action on Poisson sprinklings, $\mathrm{Tr}(f(D_N^2/\Lambda^2))$, reduces to the BD action (or a known generalization) in the appropriate regime. If it does not, identify the additional structure in the spectral action and determine whether this is a feature (richer discrete dynamics) or a bug (inconsistency with established causal set results). This is a key consistency check for Postulate 2.

Expected outcome: A rigorous (or at least well-motivated) non-perturbative definition of the SCT quantum theory, with evidence from finite-dimensional models. In particular, numerical evidence that truncated spectral triples on Poisson sprinklings converge to continuum spectral geometry.
Dependencies: MT-3 (Postulate 5 variant V3 provides the starting point), MR-1 (Lorentzian vs. Euclidean formulation).
Severity: IMPORTANT.

4.9 MR-9: Black hole singularity resolution

Task: Determine whether the nonlocal SCT equations of motion resolve the Schwarzschild singularity, producing a regular black hole geometry.
Method:

(a) Solve the linearized nonlocal field equations $G_{\mu\nu} + \Lambda^{-2}[\text{nonlocal terms}] = 0$ for a spherically symmetric static ansatz $ds^2 = -A(r)\,dt^2 + B(r)\,dr^2 + r^2\,d\Omega^2$.

(b) Determine whether the form factors F_1, F_2 produce a de Sitter core at $r \to 0$ (as in Modesto's nonlocal gravity), a bounce, or some other regular structure.

(c) Compute the Kretschmer scalar $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ and verify it remains finite everywhere.

(d) Compare with known regular black hole solutions: Hayward, Bardeen, Frolov, and the Modesto–Bambi nonlocal solution.

(e) If regular: determine the mass-radius relation for the de Sitter core and any observable signatures (modified Hawking temperature, quasi-normal modes).

(f) **Penrose singularity theorem status.** Determine which energy condition is violated by the nonlocal corrections, and whether the geodesic focusing theorem (Raychaudhuri equation) acquires defocusing terms from the SCT form factors. This explains *why* singularity resolution occurs at the level of the singularity theorems, not just in specific solutions.

Expected outcome: Either: (A) SCT resolves the singularity, providing a concrete prediction for the interior geometry and identification of which Penrose theorem premise fails, or (B) the singularity persists, indicating the $\mathcal{O}(\mathcal{R}^2)$ truncation is insufficient.
Dependencies: NT-1 (Dirac form factors), NT-1b (total SM form factors—the physical BH field equations require F_i^{total}, not just the Dirac sector), NT-2 (entire-function property—needed for singularity resolution mechanism), NT-4b (full nonlinear field equations for spherical symmetry), MT-1 (black hole entropy for consistency check).

Severity: DESIRABLE.

4.10 PPN-1: Parameterized post-Newtonian parameters

Task: Compute the full set of parameterized post-Newtonian (PPN) parameters for the nonlocal spectral action and confront with Solar System tests.
Method:

(a) **Linearized potential.** From the linearized nonlocal field equations (NT-4a), solve for the static spherically symmetric weak-field metric $ds^2 = -(1+2\Phi)\,dt^2 + (1-2\Psi)\,d\vec{x}^2$ with the nonlocal form factors. For exponential-type form factors, the Newtonian potential is

$$\Phi(r) = -\frac{Gm}{r}\,\mathrm{erf}\left(\frac{Mr}{2}\right) \tag{28}$$

where M is the mass scale set by Λ. This implies $\gamma_{\text{PPN}} \to 1$ and $\beta_{\text{PPN}} \to 1$ at distances $r \gg 1/M$ with exponentially suppressed corrections.

(b) **Full PPN formalism.** Extend to the full 10-parameter PPN framework (Will 2014). Compute all velocity-dependent and preferred-frame parameters $(\alpha_1, \alpha_2, \alpha_3, \zeta_1, \ldots)$. The spectral action is spectrally invariant (hence diffeomorphism-invariant), which should set all preferred-frame parameters to zero.

(c) **Confrontation with data:**

- Cassini bound: $|\gamma - 1| < 2.3 \times 10^{-5}$

- Lunar Laser Ranging: Nordtvedt effect, $|\eta| < 4.4 \times 10^{-4}$
- Perihelion precession of Mercury: constrains β
- Binary pulsars (Hulse–Taylor, double pulsar): radiation damping

(d) **Short-distance deviations.** Compute $\Delta F/F(r)$ at sub-millimeter scales for comparison with Eöt–Wash torsion balance experiments and atom interferometry. Determine the lower bound on Λ from null results.

Expected outcome:

- Full PPN parameter table for SCT as a function of Λ.
- Lower bound on Λ from Solar System and laboratory tests.
- Demonstration that SCT passes all current weak-field tests.
- This is the single most important calculation for experimental credibility: without PPN parameters, SCT cannot be placed in the standard framework used by experimentalists to compare gravitational theories.

Dependencies: NT-4a (linearized nonlocal field equations—the weak-field limit requires the explicit linearized equations), NT-1b (total form factors—the PPN parameters depend on F_i^{total}).

Severity: CRITICAL. This is the gold standard for testing gravitational theories.

4.11 FUND-1: Recovery of QFT on curved spacetime

Task: Demonstrate that SCT reduces to standard quantum field theory on curved spacetime in the appropriate limit.
Method:

(a) **Semiclassical limit.** Starting from the full SCT path integral (or its perturbative expansion), perform a Born–Oppenheimer-type separation of gravitational and matter degrees of freedom: fix the gravitational sector to a classical background $\bar{g}_{\mu\nu}$ and show that the matter sector reduces to QFT on $(\mathcal{M}, \bar{g}_{\mu\nu})$.

(b) **Algebraic QFT structure.** Verify that in the regime of weak and slowly varying gravitational fields, the matter algebra satisfies the Haag–Kastler axioms (isotony, locality, covariance) to the extent expected of QFT on curved spacetime.

(c) **Stress-energy renormalization.** Show that the renormalized stress-energy tensor $\langle T_{\mu\nu}\rangle_{\text{ren}}$ in the semiclassical limit reproduces the standard conformal anomaly: $\langle T^{\mu}{}_{\mu}\rangle = a\,E_4 + c\,C^2 + b\,\Box R$ with the correct central charges for the SM matter content.

(d) **Comparison:** The spectral action's heat-kernel expansion already reproduces the classical SM Lagrangian coupled to gravity. The nontrivial step is showing that the *quantum* theory on the background reproduces standard QFT including renormalization, anomalies, and running couplings.

(e) **Fermion doubling specification.** The noncommutative geometry approach to the Standard Model encounters the fermion doubling problem (Barrett 2006; Connes 2006): the real structure J on the finite spectral triple naturally produces twice the physical fermion degrees of freedom. The standard resolution (Connes–Chamseddine "first-order condition" or the "Majorana condition" on the full Hilbert space) must be: (i) explicitly stated as part of the SCT formulation, (ii) shown to be compatible with the Lorentzian spectral action (MR-1), (iii) verified to not introduce anomalies or spoil the SM quantum numbers. If the resolution changes the fermion content (as in some approaches), the effect on form factors (NT-1, NT-1b) must be assessed.

Expected outcome:

- Proof that SCT reduces to QFT on curved spacetime in the appropriate semiclassical limit.
- Explicit conditions on the gravitational background (curvature $\ll \Lambda^2$) under which this reduction is valid.
- Verification of conformal anomaly coefficients.

Dependencies: NT-1b (total SM form factors), NT-4b (full nonlinear field equations for the background), MT-3 (quantum formulation needed for semiclassical limit).

Severity: IMPORTANT. This is a fundamental consistency requirement: it connects the UV-complete theory to the regime where all existing particle physics experiments operate.

4.12 COMP-1: Cross-program comparison table

Task: Produce a detailed comparison of SCT with competing quantum gravity programs on all shared prediction channels.
Method:

(a) **Programs:** Loop quantum gravity, asymptotic safety, causal dynamical triangulations, string theory (where applicable), infinite-derivative gravity (Tomboulis–Modesto–BMS).

(b) **Comparison axes:** spectral dimension $d_S(\ell)$, BH entropy $c_{\log}$, singularity resolution mechanism, inflationary predictions (n_s, r), modified dispersion relations, PPN deviations, UV behavior (asymptotically free / safe / finite), cosmological constant treatment, matter coupling.

(c) **Distinguishing predictions.** Identify observables where SCT gives quantitatively different predictions from each competitor. The ratio c_1/c_2 is already unique; determine if there are others.

(d) **Quantitative asymptotic safety (AS) comparison.** Asymptotic safety is the closest competitor to SCT in methodology (both use the gravitational effective action with higher-curvature operators). Perform a direct, quantitative comparison:

  (i) Compare the AS fixed-point values of $g^* = G\,k^2$, $\lambda^* = \Lambda_{\rm cc}\,k^{-2}$ (Reuter et al.) with the SCT spectral action's running couplings $G(\Lambda)$, $\Lambda_{\rm cc}(\Lambda)$.

(ii) Compare the AS prediction for the spectral dimension $d_S^{\mathrm{AS}}(\ell) \to 2$ (Lauscher–Reuter 2005) with the SCT prediction from NT-3. If both give $d_S \to 2$, this is universality; if they differ, this is a distinguishing observable.

(iii) Compare the AS predictions for the graviton propagator (Codello et al. 2008) with the SCT propagator from NT-4a/MR-2. The form factor structure $F_i(z)$ from the spectral action is a concrete prediction that AS computes differently (polynomial truncation vs. nonlocal entire function).

(e) Update as each prediction task (MT-1, MT-2, INF-1, LT-3, PPN-1, NT-3) is completed.

Expected outcome:

- Publication-ready comparison table.
- Identification of the most discriminating experiments/observations between SCT and competing approaches.
- Quantitative AS vs. SCT comparison on spectral dimension, propagator, and running couplings.

Dependencies: Accumulates results from all prediction tasks. Can begin with existing results and be updated iteratively.

Severity: DESIRABLE. Important for positioning SCT in the field, but not a theory-internal consistency requirement.

5 Long-Term Tasks

These are the ultimate goals of the SCT program. Each requires substantial results from mid-term and consistency tasks.

5.1 LT-1: UV-completeness

Task: Prove the existence of a fixed-point structure (asymptotic freedom, asymptotic safety, or finiteness) for the full nonlocal spectral action functional under the renormalization group.

Method:

(a) Define the RG flow for the spectral action. The natural cutoff is Λ appearing in (1). Define the effective action at scale k:

$$\Gamma_k = \mathrm{Tr}\left(f_k\left(\frac{D^2}{k^2}\right)\right) \tag{29}$$

where f_k flows with the Wetterich–Morris exact RG equation.

(b) Compute the beta functions for the couplings encoded in f_k. In the truncation to local operators, these include $G(k)$, $\Lambda_{\mathrm{cc}}(k)$, $c_1(k)$, $c_2(k)$.

(c) Search for fixed points: $\beta_i(g^*) = 0$ for all couplings.

(d) If the entire-function property (NT-2) holds, investigate whether the nonlocal structure is preserved under RG flow ("nonlocal asymptotic safety").

(e) Prove or disprove finiteness: if the spectral action is UV-finite at all loop orders, this would establish SCT as a UV-complete theory without requiring a fixed point.

(f) **UV-completeness criterion clarification.** Before attempting the proof, establish what "UV-complete" means for SCT specifically. The three possible outcomes are:

   (i) **Asymptotic freedom** ($g_i(k) \to 0$ as $k \to \infty$): the theory is weakly coupled in the UV. Requires identification of the relevant couplings and their one-loop beta functions.

   (ii) **Asymptotic safety** ($g_i(k) \to g_i^*$ as $k \to \infty$, $g_i^* \neq 0$ with finitely many relevant directions): the theory has an interacting UV fixed point with finite predictive power.

   (iii) **UV finiteness**: all scattering amplitudes are finite at all orders without renormalization. This is the strongest claim and would require proof that the nonlocal form factors F_i suppress all UV divergences.

   The criterion must be stated *before* the calculation, not chosen post-hoc to match whatever is found.

Expected outcome:

- Proof (or strong evidence) for UV-completeness of SCT.
- Classification: asymptotically free / safe / finite / none.
- If none: identification of what additional structure is needed.
- Explicit criterion for UV-completeness stated in advance of the calculation, with clear success/failure conditions.

Dependencies: NT-1 (Dirac form factors), NT-1b (total SM form factors—the RG beta functions $\beta(G)$, $\beta(c_1)$, $\beta(c_2)$ depend on the full matter content), NT-2 (analyticity structure), MT-3 (correct formulation of quantum theory), MR-2 (unitarity and stability—UV-completeness requires the theory to be ghost-free), MR-4 (two-loop calculation—determines subleading UV structure), MR-5 (all-orders finiteness argument).

5.2 LT-2: Information paradox and island formula

Task: Derive the island formula for entanglement entropy from SCT principles, and resolve the black hole information paradox within the spectral framework.
Method:

(a) Start from the entanglement entropy of the spectral triple (MT-1).

(b) Derive the quantum extremal surface (QES) condition from the spectral action:

$$\frac{\delta}{\delta X^\mu}\left[\frac{\mathrm{Area}(\gamma_A)}{4G_{\mathrm{eff}}} + S_{\mathrm{bulk}}(\Sigma_A)\right] = 0 \tag{30}$$

where G_{eff} is the effective Newton constant from the spectral action and S_{bulk} is the bulk entanglement entropy.

(c) Show that the Page curve emerges from the competition between "no-island" and "island" saddles in the spectral path integral.

(d) Determine whether the spectral triple naturally implements the "code" model: $S(A) = \mathrm{Area}(\gamma_A)/(4G_{\mathrm{eff}}) + S_{\mathrm{bulk}}$ as a *definition* of geometry from entanglement data.

Expected outcome:

- Island formula derived from spectral principles (not postulated).
- Explicit Page curve for evaporating black hole in SCT.
- Understanding of how information is preserved in the spectral framework.
- **Firewall resolution.** SCT's inherent nonlocality provides a natural resolution mechanism (Giddings-type nonlocal information transfer): the form factors spread quantum effects over scales $\sim 1/\Lambda$ near the horizon, potentially smoothing the firewall without violating monogamy of entanglement. Make this connection explicit and determine whether the AMPS paradox is resolved or merely reframed.

Dependencies: MT-1 (black hole entropy), MT-3 (quantum formulation), LT-1 (UV-completeness needed for late-time behavior).

5.3 LT-3: Testable numerical predictions

The original single task is split into four focused sub-tasks, each targeting a distinct experimental channel with its own methodology and dataset.

5.3.1 LT-3a: Quasinormal mode shifts

Task: Compute the QNM frequency shifts for Schwarzschild and Kerr black holes from the nonlocal SCT action and confront with gravitational wave ringdown observations.
Method:

(a) From the linearized nonlocal field equations (NT-4b) on a Schwarzschild background, derive the modified Regge–Wheeler and Zerilli equations with nonlocal potentials depending on $F_1(\Box/\Lambda^2)$, $F_2(\Box/\Lambda^2)$.

(b) Compute the QNM frequencies:

$$\omega_{\mathrm{QNM}}^{\mathrm{SCT}} = \omega_{\mathrm{QNM}}^{\mathrm{GR}} + \delta\omega\left(\frac{c_1}{\Lambda^2}, \frac{c_2}{\Lambda^2}\right) \tag{31}$$

with the ratio c_1/c_2 fixed by the spectral triple (NT-1b step e). The shifts $\delta\omega$ depend on a single parameter Λ.

(c) Extend to Kerr using the modified Teukolsky equation.

(d) Compare with LVK O4/O5 ringdown data; project sensitivity for LISA and Einstein Telescope (ET).

Expected outcome: QNM frequency table as function of Λ; bound on Λ from current ringdown observations; forecast for next-generation detectors.
Dependencies: NT-1b, NT-4b (nonlinear equations on BH background), MT-1 (BH background), MR-9 (for near-singularity QNMs).

5.3.2 LT-3b: Gravitational wave propagation

Task: Compute the GW speed, polarization content, and frequency-dependent dispersion from the nonlocal SCT action.
Method:

(a) **GW speed.** From the linearized equations (NT-4a), derive the group and phase velocities of gravitational waves on an FLRW background. Ghost-free infinite-derivative gravity predicts $c_{\rm GW} = c$ (Buoninfante et al. 2019), consistent with the GW170817 bound $|c_{\rm GW}/c - 1| < 10^{-15}$. Verify this for SCT's specific form factors.

(b) **Polarization.** Determine the number of propagating polarization modes. Ghost-free IDG gives only 2 tensor modes (same as GR); verify this persists in the full nonlocal theory with SCT-fixed coefficients.

(c) **Dispersion.** The nonlocal form factors introduce frequency-dependent corrections to the dispersion relation: $\omega^2 = k^2 + \delta\omega^2(k;\,\Lambda)$. Compute $\delta\omega^2(k)$ and constrain using multi-messenger observations (GW170817/GRB 170817A), Fermi-LAT/GBM time-of-flight bounds on Lorentz-violating dispersion, and IceCube/KM3NeT high-energy neutrino timing constraints.

(d) **Secondary GW spectrum.** If INF-1 produces enhanced small-scale perturbations, compute the induced secondary gravitational wave background. Compare with PTA observations (NANOGrav, EPTA, PPTA).

Expected outcome: GW propagation properties of SCT; bounds from multi-messenger astronomy; compatibility with existing constraints.
Dependencies: NT-4a (linearized propagation equations), NT-4c (FLRW background), INF-1 (for secondary spectrum).

5.3.3 LT-3c: Cosmological predictions

Task: Produce numerical predictions for the cosmological equation of state $w(z)$ and structure formation observables from the SCT-modified Friedmann equations.
Method:

(a) From MT-2 (modified Friedmann equations), compute:

$$w(z) = -1 + \delta w(z;\,\Lambda) \tag{32}$$

Fit to the CPL parametrization $w(z) = w_0 + w_a\, z/(1+z)$.

(b) Implement SCT modifications in CLASS/hi_class + cobaya. Compute the CMB temperature and polarization power spectra C_ℓ^{TT}, C_ℓ^{EE}, C_ℓ^{TE} and the matter power spectrum $P(k)$.

(c) Run MCMC fits against: Planck 2018 + ACT DR6 (CMB), DESI DR1/DR2 (BAO), Pantheon+ / Union3 (SNe Ia), KiDS-1000/DES-Y3 (weak lensing). Report constraints on Λ and comparison with ΛCDM (Bayesian evidence, $\Delta\chi^2$, AIC/BIC).

(d) Project sensitivity of Euclid, LSST (Vera Rubin), CMB-S4, Simons Observatory ($\sigma_r \approx 0.003$).

Expected outcome: Constraints on Λ from cosmological data; model comparison with ΛCDM; forecast for next-generation surveys.
Dependencies: MT-2, NT-1b, NT-4c (FLRW reduction for cosmological perturbations).

5.3.4 LT-3d: Laboratory and Solar System tests

Task: Compute the short-distance modification of gravity and confront with laboratory and Solar System experiments.
Method:

(a) The nonlocal form factors modify the Newtonian potential at short distances:

$$V(r) = -\frac{Gm_1m_2}{r}\left(1 + \frac{\Delta F}{F}(r;\,\Lambda)\right) \tag{33}$$

Compute $\Delta F/F(r)$ from the exact form factors (NT-1b, NT-4a).

(b) Compare with torsion balance experiments (Eöt–Wash group: $\alpha < 10^{-2}$ at $r \sim 50\ \mu$m), atom interferometry, and Casimir force measurements.

(c) Combine with PPN-1 results for Solar System constraints.

(d) Produce a unified exclusion plot in the (Λ, α) plane combining all experimental channels.

Expected outcome: Lower bound on Λ from laboratory experiments; unified exclusion plot; identification of the most sensitive experiment for testing SCT at each distance scale.
Dependencies: NT-1b, NT-4a (linearized equations for short-distance potential), PPN-1 (Solar System).

5.3.5 LT-3e: Stellar Structure and TOV limits

Task: Derive the modified Tolman–Oppenheimer–Volkoff (TOV) equations for SCT and compute the maximum mass of neutron stars.
Method:

(a) Set up the interior static, spherically symmetric metric with a perfect fluid source.

(b) Treat the non-local higher-derivative terms perturbatively in $1/\Lambda^2$ (or use auxiliary scalar fields) to bypass the intractability of exact non-local integro-differential equations inside matter.

(c) Integrate the modified TOV equations for realistic nuclear equations of state (e.g., SLy, AP4).

(d) Verify that the non-local form factors do not artificially lower the maximum mass below the observational limit of $\sim 2.14 M_\odot$.

Expected outcome: Maximum neutron star mass as a function of Λ; theoretical viability check against heavy pulsar observations.
Dependencies: NT-4b (full nonlinear equations are required for strong-field interior metrics).
Severity: CRITICAL. (Direct astrophysics falsification point).

6 Lessons from the Analysis of the Combined Approach (Part IV)

The analysis of a combined approach (referred to as the "stitched" model, analogous to an engineering assembly of foreign modules) in Part IV yielded both cautionary lessons and constructive insights for SCT.

6.1 Structural critique of the stitched approach

The combined model assembles three independent mechanisms:

1. **Vacuum energy sequestration** (Kaloper–Padilla type): a global constraint decouples the cosmological constant from UV physics.
2. **Ghost-free nonlocality** (Tomboulis–Modesto–Biswas–Mazumdar–Siegel type): entire-function form factors tame UV divergences.
3. **Islands/QES** (Ryu–Takayanagi–Engelhardt–Wall type): the island formula for entanglement entropy.

Key deficiencies identified:

- **No single generating principle.** The three sectors are simply added, with no explanation of why these particular mechanisms coexist or are compatible.
- **UV-completion not specified.** The model explicitly marks UV-completion as an open question, offering no path forward.
- **Spectral dimension $d_S : 4 \to 2$ not mentioned.** A universal feature of quantum gravity approaches is absent.
- **Black hole entropy not derived.** The Bekenstein–Hawking formula is assumed, not obtained from the framework.
- **No new predictions beyond component models.** The combination does not generate predictions that its components individually lack.

6.2 Strategic contrast: SCT vs. stitched approach

Feature	**Stitched model**	**SCT**
Generating principle	Three separate	One (spectral triple)
Form factors	Postulated	Must be derived
Sequestration	Added	Potentially automatic
BH entropy	Assumed	Must be derived
d_S flow	Not addressed	Must be computed
UV-completion	"Not specified"	Must be proven

6.3 Constructive lessons for SCT

Despite structural deficiencies, the stitched model identifies useful directions:

1. **Prediction format.** The three channels—smoothed Newtonian potential $\Delta F/F(r)$, QNM frequency shifts $\delta\omega$, and cosmological equation of state $w(z)$—are the correct prediction targets. SCT should produce numbers in these same channels.

2. **Entire form factors as effective description.** The postulated ghost-free form factors work well as a phenomenological model. In SCT, the key question is whether these arise *naturally* from the spectral triple (Task NT-2).

3. **Sequestration as a hint.** The spectral action $\mathrm{Tr}(f(D^2/\Lambda^2))$ may be automatically insensitive to additive shifts in the vacuum energy, providing a natural sequestration mechanism without a separate postulate. This should be investigated.

4. **The "code" model.** The relation $S(A) = \mathrm{Area}(\gamma_A)/(4G_{\mathrm{eff}}) + S_{\mathrm{bulk}}$ as a *definition* of geometry from entanglement data is closest to the SCT spirit, where geometry is spectral data. Exploring this connection is a priority for Task LT-2.

7 Priority Order and Dependencies

7.1 Dependency graph

The tasks form the following dependency structure (arrows indicate "must be completed before"):

Level 0 (Foundation — proven results):

Spectral action $\to$ Einstein–Hilbert limit [Eq. (2)]
Ratio $c_1/c_2 = -1/3$ [Eq. (4)]
Master function $f(\xi)$ [Eq. (5)]

Level 1 (Near-term + independent):

NT-1 (Dirac form factors) $\longleftarrow$ Level 0 [COMPLETE]
NT-1b (Scalar + vector form factors) $\longleftarrow$ NT-1 [PHASES 1–2 COMPLETE]
NT-2 (Entire-function property) $\longleftarrow$ NT-1
NT-3 (Spectral dimension) $\longleftarrow$ Level 0 (partial); MT-3 (full)
NT-4a (Linearized nonlocal eqs.) $\longleftarrow$ NT-1, NT-1b
META-1 (Parameter counting + f-analysis) $\longleftarrow$ Level 0 (iterative)
MR-1 (Lorentzian spectral action) $\longleftarrow$ Level 0 [CRITICAL]
MR-6 (Curvature expansion convergence) $\longleftarrow$ Level 0

Level 1.5 (NT-4 sub-tasks):

NT-4b (Full nonlinear eqs.) $\longleftarrow$ NT-4a, NT-1, NT-1b
NT-4c (FLRW reduction) $\longleftarrow$ NT-4b, NT-1b

Level 2 (Mid-term and consistency):

MT-1 (BH entropy + all 4 laws) $\longleftarrow$ NT-1, NT-1b, NT-4b
MT-2 (Cosmology + H_0/S_8) $\longleftarrow$ NT-1, NT-1b, NT-2, NT-4c
MT-3 (Postulate 5 variants) $\longleftarrow$ NT-1, NT-1b, NT-2
INF-1 (Inflation: n_s, r, f_{NL}) $\longleftarrow$ NT-1b, NT-2, NT-4c
PPN-1 (PPN parameters) $\longleftarrow$ NT-1b, NT-4a [CRITICAL]
FUND-1 (QFT on curved spacetime) $\longleftarrow$ NT-1b, NT-4b, MT-3
MR-2 (Unitarity + stability) $\longleftarrow$ NT-1, NT-1b, NT-2, NT-4a [CRITICAL]
MR-3 (Causality + macrocausality) $\longleftarrow$ NT-1, NT-1b, NT-2, NT-4a [CRITICAL]
MR-7 (Scattering amplitudes) $\longleftarrow$ NT-1b, NT-4a, MR-2, MR-3
MR-9 (BH singularity + Penrose) $\longleftarrow$ NT-1, NT-1b, NT-2, NT-4b, MT-1
COMP-1 (Cross-program comparison) $\longleftarrow$ (iterative, all prediction tasks)

Level 3 (Long-term):

MR-4 (Two-loop) $\longleftarrow$ NT-1, NT-1b, MR-2, MR-7 [CRITICAL]
MR-5 (All-orders + naturalness) $\longleftarrow$ NT-1, NT-1b, NT-2, MR-4 [CRITICAL]
MR-8 (Non-perturbative definition) $\longleftarrow$ MT-3, MR-1
LT-1 (UV-completeness) $\longleftarrow$ NT-1, NT-1b, NT-2, MT-3, MR-2, MR-4, MR-5
LT-2 (Information paradox + firewalls) $\longleftarrow$ MT-1, MT-3, LT-1
LT-3a (QNM predictions) $\longleftarrow$ NT-1b, NT-4b, MT-1, MR-9
LT-3b (GW propagation) $\longleftarrow$ NT-4a, NT-4c, INF-1
LT-3c (Cosmological predictions) $\longleftarrow$ MT-2, NT-1b, NT-4c
LT-3d (Laboratory/Solar System) $\longleftarrow$ NT-1b, NT-4a, PPN-1

7.2 Critical path

The critical path (longest chain of dependent tasks) for UV-completeness is:

$$\boxed{\text{NT-1}\checkmark} \longrightarrow \boxed{\text{NT-1b}} \longrightarrow \boxed{\text{NT-4a}} \longrightarrow \boxed{\text{MR-2}} \longrightarrow \boxed{\text{MR-7}} \longrightarrow \boxed{\text{MR-4}} \longrightarrow \boxed{\text{MR-5}} \longrightarrow \boxed{\text{LT-1}}$$

(NT-2, also required by MR-2, runs in parallel with NT-1b $\rightarrow$ NT-4a and does not lengthen this path.)

A parallel critical path for the physical predictions track (cosmological):

$$\boxed{\text{NT-1}\checkmark} \longrightarrow \boxed{\text{NT-1b (Ph. 1–2}\checkmark)} \longrightarrow \boxed{\text{NT-4a}} \longrightarrow \boxed{\text{NT-4b}} \longrightarrow \boxed{\text{NT-4c}} \longrightarrow \boxed{\text{INF-1}} \longrightarrow \boxed{\text{LT-3}}$$

A third critical path for Solar System/laboratory tests (shortest):

$$\boxed{\text{NT-1}\checkmark} \longrightarrow \boxed{\text{NT-1b}} \longrightarrow \boxed{\text{NT-4a}} \longrightarrow \boxed{\text{PPN-1}} \longrightarrow \boxed{\text{LT-3d}}$$

Key advantage of the NT-4 split: The Solar System path now requires only NT-4a (linearized equations), not the full NT-4b/c. This de-risks the most experimentally urgent critical path.

NT-1 is COMPLETE. NT-1b Phases 1–2 (scalar and vector form factors) are COMPLETE; Phase 3 (c_1/c_2 re-derivation with full SM spectrum) remains **pending**. Once

Phase 3 finishes, NT-1b unblocks NT-4a (linearized equations), MR-2 (propagator), and MR-7 (amplitudes). After NT-1b, **NT-4a (linearized nonlocal equations)** feeds directly into PPN-1, MR-2, MR-3, and MR-7, while **NT-4b** (full nonlinear) feeds into MT-1, MR-9, and **NT-4c** (FLRW reduction) feeds into MT-2, INF-1. In parallel, **MR-1 (Lorentzian spectral action)** and **MR-6 (convergence)** can proceed independently.

7.3 Recommended execution order

Pri.	Task	Blocked by	Unblocks
—	NT-1: Dirac form factors	DONE	(all below)
1	META-1: Theory census + f-analysis	—	(foundation for all reviews)
2	NT-1b: Ph. 1–2 done; Ph. 3 (c_1/c_2) pending	NT-1	NT-4a, MR-2, MR-4, MR-7, MT-1, MT-2, INF-1, PPN-1, LT-3a–e
3	MR-1: Lorentzian action	—	MR-8
4	MR-6: Convergence	—	(cross-check)
5	NT-2: Entire functions	NT-1	MR-2, MR-3, MT-2, MT-3, MR-5, INF-1
6	NT-3: Spectral dimension	— (partial)	(interpretation)
7a	NT-4a: Linearized nonlocal eqs.	NT-1, NT-1b	MR-2, MR-3, MR-7, PPN-1, NT-4b
7b	NT-4b: Full nonlinear eqs.	NT-4a	MT-1, MR-9, NT-4c, FUND-1, LT-3a
7c	NT-4c: FLRW reduction	NT-4b, NT-1b	MT-2, INF-1, LT-3b, LT-3c
8	MR-2: Unitarity+stability	NT-1b, NT-2, NT-4a	MR-4, MR-7, LT-1
9	MR-3: Causality+macrocausality	NT-1b, NT-2, NT-4a	MR-7
10	PPN-1: Post-Newtonian limits	NT-4a, NT-1b	LT-3d (verification)
11	MT-2: Cosmology+tensions	NT-1b, NT-2, NT-4c	LT-3b, LT-3c
12	MT-1: BH entropy+4 laws	NT-1b, NT-4b	LT-2, LT-3a, MR-9
13	INF-1: Spectral inflation	NT-1b, NT-2, NT-4c	LT-3b, LT-3c
14	MT-3: Postulate 5	NT-1, NT-1b, NT-2	LT-1, LT-2, MR-8, FUND-1
15	MR-7: Scattering ampl.	NT-1b, NT-4a, MR-2, MR-3	MR-4
16	MR-9: BH singularity+Penrose	NT-1b, NT-2, NT-4b, MT-1	LT-3a
17	FUND-1: Axiomatic foundations	NT-1b, NT-4b, MT-3	(structural integrity)
18	MR-4: Two-loop	NT-1, NT-1b, MR-2, MR-7	MR-5, LT-1

Pri.	Task	Blocked by	Unblocks
19	MR-5: All-orders+naturalness	NT-1, NT-1b, NT-2, MR-4	LT-1
20	MR-8: Non-perturbative	MT-3, MR-1	(structural integrity)
21	LT-3a: QNM predictions	NT-1b, NT-4b, MT-1, MR-9	(output)
22	LT-3b: GW propagation	NT-4a, NT-4c, INF-1	(output)
23	LT-3c: Cosmological predictions	MT-2, NT-1b, NT-4c	(output)
24	LT-3d: Laboratory/Solar System	NT-1b, NT-4a, PPN-1	(output)
25	LT-3e: Stellar structure (TOV)	NT-4b	(output)
26	LT-1: UV-completeness	NT-1b, NT-2, MT-3, MR-2, MR-4, MR-5	LT-2
27	LT-2: Info. paradox+firewalls	MT-1, MT-3, LT-1	(final output)

COMP-1 (cross-program comparison) runs iteratively throughout, updated as results accumulate.

7.4 Parallelization opportunities

The following tasks can proceed simultaneously:

- **Now (NT-1 complete):** NT-1b, NT-2, MR-1, MR-6, NT-3, **META-1** are all independent of each other. META-1 (theory census + cutoff function analysis) has no dependencies and should start immediately.
- **After NT-1b:** NT-4a (linearized equations) can start (needs only NT-1 + NT-1b).
- **After NT-4a: PPN-1** (post-Newtonian limits) can start immediately, in parallel with MR-2, MR-3, and NT-4b. PPN-1 is **CRITICAL**—a Solar System test of the linearized potential. *Key advantage of the NT-4 split:* PPN-1 no longer waits for the full nonlinear variational problem (NT-4b).
- **After NT-4a + NT-2:** MR-2 and MR-3 can proceed in parallel. MR-7 follows once MR-2 + MR-3 are both complete (depends on both).
- **After NT-4b:** NT-4c (FLRW reduction) starts, leading to MT-2 and INF-1.
- **After NT-4c:** MT-2 and INF-1 can proceed in parallel (INF-1 no longer depends on MT-2).
- **MT-1 and MT-2** are independent (MT-1 needs NT-4b; MT-2 needs NT-4c).
- **MR-7 → MR-4:** MR-7 (scattering amplitudes) must now precede MR-4 (two-loop), as the one-loop integrand structure feeds the two-loop calculation.
- **LT-3a/b/c/d** are largely independent of each other: LT-3a (QNMs) needs MT-1; LT-3b (GW propagation) needs INF-1; LT-3c (cosmology) needs MT-2; LT-3d (lab tests) needs PPN-1.

- **LT-1 and LT-3a–d** are independent (LT-3 does not require LT-1).
- **MR-8 and MR-4** are independent (different dependency chains).
- **MR-9** can proceed as soon as NT-4b + MT-1 are done, independently of the UV-completeness track.
- **FUND-1** can proceed after NT-4b + MT-3, independently of prediction tasks.
- **COMP-1** runs iteratively—updated after each major result, no blocking dependencies.

7.5 Methods and tools per task

Task	Primary tools	Verification
NT-1	SymPy, SciPy, `sct_tools`	DONE. 274+ checks PASS (7 audit rounds).
NT-1b	SymPy, SciPy, `sct_tools`, cadabra2^{W}	Local limits vs. known β-functions; consistency with NT-1 Dirac results
NT-2	Complex analysis (SymPy), mpmath	Hadamard factorization; numerical pole search
NT-3	`sct_tools.graphs`, networkx, SciPy	Check $d_S \to 4$ in IR; compare with CDT lattice results
NT-4a	SymPy (linearized), `sct_tools`	Flat-space limit $\to$ massless graviton; gauge invariance check
NT-4b	cadabra2^{W} (variational)	Local limit $\to$ Einstein eqs.; Bianchi identity; trace consistency
NT-4c	SymPy, SciPy (FRW reduction)	de Sitter limit; GR Friedmann recovery; perturbation stability
META-1	Manual analysis, LaTeX tables	10-check review; competitor comparison; falsifiability matrix
MR-1	Formal (Krein space theory)	Wick rotation consistency; compare Paschke–Sitarz, van den Dungen
MR-2	SymPy, cadabra2^{W} (propagator)	Residue signs; optical theorem; comparison with Stelle, Tomboulis
MR-3	SymPy + SciPy (Green's fn.)	Kramers–Kronig; retarded propagator; macrocausality (Krasnikov test)
MR-4	cadabra2^{W}, pySecDec^{W} (2-loop)	Compare Goroff–Sagnotti; cross-check divergence cancellation
MR-5	Formal (power-counting)	Explicit check at 2-loop (MR-4); compare Tomboulis, Modesto arguments
MR-6	SymPy + numerical	Exact S^4 spectral action vs. truncated expansion
MR-7	cadabra2^{W}, SymPy (amplitudes)	Ward identities; IR limit = GR; unitarity cuts

Task	Primary tools	Verification
MR-8	Formal + numerical (matrix models)	Barrett–Glaser comparison; CDT cross-check; Poisson sprinkling convergence (Glaser–Stern)
MR-9	SciPy (ODE integration)	Kretschmer scalar finiteness; Penrose theorem; Raychaudhuri defocusing
PPN-1	SymPy (linearized eqs.), SciPy	Cassini bound; LLR; Mercury perihelion; binary pulsars; Eöt–Wash
INF-1	SymPy, CAMB, CLASS^{W}, cobaya	Planck n_s/r; BICEP/Keck; LiteBIRD forecasts; non-Gaussianity
FUND-1	Formal analysis, SymPy	Born–Oppenheimer sep.; Haag–Kastler axioms; anomaly check
COMP-1	Literature analysis, LaTeX tables	Feature-by-feature comparison; domain map vs. 5 programs
MT-1	SymPy (Wald entropy), healpy^{W}	Compare with Bekenstein–Hawking; LQG $c_{\log}$; 4 laws
MT-2	CAMB, CLASS^{W}, healpy^{W}, emcee	Fit to Planck + DESI + Pantheon+; H_0 and S_8 tensions; Λ_{CC} consistency (spectral vs. causal set)
MT-3	Formal analysis, SymPy	Well-posedness proofs; classical limit check
LT-1	Functional RG (numerical), pySecDec^{W}	Fixed-point search; requires MR-2, MR-4, MR-5 inputs
LT-2	Formal + numerical	Page curve; firewall resolution (Giddings-type); unitarity check
LT-3a	SciPy (ODE), cadabra2^{W}	Modified Regge–Wheeler/Zerilli; Kerr via Teukolsky; LIGO/ET data
LT-3b	SymPy, SciPy (dispersion)	GW speed; polarization; secondary spectrum; LISA forecasts
LT-3c	CLASS^{W}, cobaya, healpy^{W} (MCMC)	$w(z)$ from hi_class; DESI + Euclid comparison
LT-3d	SciPy (numerical), SymPy	Eöt–Wash torsion balance; Casimir; unified exclusion plot
LT-3e	SciPy (ODE integration)	Max neutron star mass $\geq 2.14 M_\odot$

^{W} Runs in WSL2 Ubuntu 24.04 (venv `~/sct-wsl`). cadabra2: tensor CAS (index contractions, Bianchi identities, variational calculus). pySecDec: multi-loop Feynman integrals via sector decomposition. healpy: HEALPix CMB analysis (angular power spectra, C_ℓ estimation). CLASS: Boltzmann solver (alternative to CAMB for cross-validation).

Summary

Completed: NT-1 (Dirac form factors F_1, F_2) and NT-1b Phases 1–2 (scalar + vector form factors, β_W, β_R for all spins) are complete with 274+142+1519 = 1935+ quantitative checks passing across 7 independent audit rounds.

Current status: The roadmap contains **31 tasks** (27 original, with NT-4 split adding 2, LT-3e adding 1; plus COMP-1 running iteratively) organized in five categories:

- **Near-term (5+2 sub-tasks):** NT-1 [DONE], NT-1b [Ph. 1–2 DONE] (Ph. 3: c_1/c_2 pending), NT-2, NT-3, NT-4a (linearized), NT-4b (full nonlinear), NT-4c (FLRW reduction).
- **Mid-term (3+2):** MT-1 (BH entropy + 4 laws), MT-2 (cosmology + tensions), MT-3 (Postulate 5); plus **INF-1** (spectral inflation) and **META-1** (theory census + cutoff function f analysis).
- **Theory consistency (9+2):** MR-1 through MR-9; plus **PPN-1** (post-Newtonian limits, **CRITICAL**) and **FUND-1** (axiomatic foundations + fermion doubling). MR-1, MR-2, MR-3, MR-4, MR-5, and PPN-1 are **CRITICAL** for theory viability.
- **Long-term (7):** LT-1 (UV-completeness), LT-2 (information paradox + firewalls), LT-3a (QNMs), LT-3b (GW propagation), LT-3c (cosmological predictions), LT-3d (laboratory/Solar System tests), LT-3e (stellar structure/TOV limits).
- **Iterative:** COMP-1 (cross-program comparison + quantitative AS comparison, updated continuously).

Immediate priorities (can proceed in parallel):

1. **META-1**: Theory census, falsifiability matrix, and cutoff function f sensitivity analysis (no dependencies; foundational self-audit).
2. **NT-1b Phase 3**: Re-derive c_1/c_2 with full SM spectrum and corrected sign $E = -R/4$ (Phases 1–2 done; unblocks NT-4a, MR-2, MR-4, MR-7, MT-1, MT-2, INF-1, PPN-1).
3. **NT-2**: Entire-function property (unblocks MR-2, MR-3, MR-5, MT-2, MT-3, INF-1).
4. **MR-1**: Define the Lorentzian spectral action (foundational; independent of all other tasks).
5. **MR-6**: Convergence of curvature expansion (foundational; independent).
6. **NT-3**: Spectral dimension flow (partially independent).

High-priority after NT-4a (linearized equations):

- **PPN-1 (CRITICAL):** Solar System tests are the gold standard for any gravity theory. The Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ is the tightest metric theory constraint. Failure here would be fatal. *Key advantage of NT-4 split:* PPN-1 needs only NT-4a (linearized), not the full nonlinear equations.

Three critical paths:

1. **UV-completeness:** NT-1✓ → NT-1b → NT-4a → MR-2 → MR-7 → MR-4 → MR-5 → LT-1 (NT-2 parallel with NT-1b→NT-4a)
2. **Cosmological predictions:** NT-1✓ → NT-1b → NT-4a → NT-4b → NT-4c → INF-1 → LT-3b (LT-3c shares NT-4c but reaches it via MT-2, not INF-1)

3. **Solar System tests (shortest):** NT-1✓ → NT-1b → NT-4a → PPN-1 → LT-3d

Key datasets:

- CMB: Planck 2018, ACT DR6, Simons Observatory (upcoming), LiteBIRD (upcoming)
- BAO: DESI DR1/DR2, Euclid (upcoming)
- SNe: Pantheon+, Union3
- GW: LVK O4/O5 (ringdown, speed), LISA (upcoming), Einstein Telescope (upcoming)
- Dispersion: Fermi GBM, IceCube/KM3NeT (neutrino timing)
- Weak lensing: KiDS-1000, DES-Y3, HSC
- CMB r: BICEP/Keck, Simons Observatory, LiteBIRD
- PPN: Cassini, LLR, Mercury perihelion, binary pulsars, Eöt-Wash

The strategic advantage of SCT over combined/stitched approaches lies in its *single generating principle*: all structure—form factors, ghost-freedom, entropy, dimensional flow—must emerge from one object, the spectral triple. The 11 theory-consistency tasks (MR-1 through MR-9, PPN-1, FUND-1) are specifically designed to test whether this claim survives the requirements for a genuine UV-complete theory. Any single CRITICAL task failure would require fundamental revision of the theory.